When Do Amplitude-Encoding Quantum Kernels Help Security Anomaly Detection? A Matched-Scale Study on QKD Telemetry and Network Intrusion Data

Arnav Kapoor

***Abstract*—Amplitude encoding is the most common way of loading classical data into quantum kernels proposed for security tasks. Its fidelity kernel is exactly the squared cosine similarity of the inputs, so it can be evaluated at any scale and compared fairly with classical kernels. We do so on four datasets in two domains—QKD physical-layer telemetry from a decoy-state BB84 simulator with operating drift, and zero-trust network flows from a controlled simulator, UNSW-NB15 and CIC-IDS2017—training every model on the same data at every training size. In distribution, the amplitude-encoding SVM is never better than an RBF-kernel SVM (e.g. 0.862 vs. 0.863 accuracy on UNSW-NB15), needs more data at small n, and trails tree ensembles and neural networks. On attack types withheld from training the picture is domain-dependent. In both controlled settings where unseen attacks differ from normal traffic mainly in magnitude, it fails (mean held-out AUC 0.53 vs. 0.84 for RBF on synthetic flows; 0.55 vs. 0.94 on QKD telemetry without its one magnitude-carrying feature); a norm-anchored encoding that appends one constant coordinate repairs this (0.79, 0.94), and interventions confirm the kernel's norm invariance as the cause. On the two public benchmarks it does not fail: it matches or slightly exceeds RBF (0.95 vs. 0.93 on UNSW-NB15; 0.71 vs. 0.66 on CIC-IDS2017), though both kernels trail tree ensembles and a benign-only autoencoder on CIC-IDS2017. A simple norm-separability statistic does not predict which case a dataset falls into, so we recommend testing unseen-attack generalization directly. We also find that a Gaussian-process posterior is a worse basis for alert deferral than an isotonic-calibrated random forest, and that drift-free QKD simulators are trivially separable. Code, data preparation and all result files are released.**

***Index Terms*—Quantum machine learning, quantum kernels, amplitude encoding, intrusion detection, quantum key distribution, zero-day attacks, calibration, negative results.**

I. Introduction

QUANTUM kernel methods map data into the Hilbert space of a quantum register and use state overlaps as a kernel [1], [2]. They are increasingly proposed for security tasks, from intrusion detection [3] to botnet traffic classification [4], usually on the strength of their novelty and of encouraging accuracy at a single operating point. Two facts make such results hard to interpret. First, whether a quantum kernel helps depends on how its geometry matches the data [5]; provable advantages exist only for specially constructed problems [6]. Second, quantum kernels are often trained on far fewer samples than the classical baselines they are compared with, because circuit simulation is slow, so model and data size are confounded.

This paper studies the most common encoding in this literature, amplitude encoding, whose fidelity kernel is exactly the squared cosine similarity of the inputs (Proposition 1). Because it is classically computable, we can evaluate it at any scale, against classical kernels trained on identical data, and ask a question that does not depend on quantum hardware: *is its inductive bias useful for security anomaly detection, and if so when?*

We answer it in two structurally different domains. The first is physical-layer telemetry from quantum key distribution (QKD) links, where implementation attacks such as photon-number splitting [7], detector blinding [8], Trojan-horse probing [9] and intercept-resend [10] perturb channel statistics. The second is network-flow monitoring for zero-trust architectures [11], [12], evaluated on a controlled flow simulator and on two public benchmarks, UNSW-NB15 [13] and CIC-IDS2017 [14].

Contributions.

1) **A matched-scale comparison** of the amplitude-encoding kernel with RBF, Laplacian and polynomial kernels, tree ensembles and neural networks, all trained on identical data at every training size, across four datasets in two security domains (Section VI-A–VI-B).
2) **A domain-dependent zero-day effect, its mechanism and a fix.** The kernel fails on unseen attack types in two controlled settings and not on two public benchmarks. Interventions—removing a magnitude-carrying feature, removing sign ambiguity, and a one-coordinate norm anchor—identify norm invariance as the cause where it fails (Sections III, VI-C, VI-D).
3) **Evidence on uncertainty for alert deferral:** a Gaussian-process posterior is less calibrated, and supports a worse coverage–accuracy trade-off, than an isotonic-calibrated random forest (Section VI-E).
4) **A QKD telemetry simulator with leakage tests** and an attack-strength axis, motivated by our finding that a drift-free simulator is trivially separable (Section V-A).

Section II reviews related work. Section III analyses the kernel. Section IV gives the evaluation protocol, Section V the four datasets, and Section VI the results. Section VII discusses implications and threats to validity.

A. Kapoor is with the Department of Computer Science, Indian Institute of Science Education and Research Bhopal, Bhopal 462066, India (e-mail: arnavkapoor23@iiserb.ac.in).

II. Background and Related Work

A. Quantum and quantum-inspired kernels

A quantum kernel evaluates $k(\mathbf{x}, \mathbf{y}) = |\langle\psi(\mathbf{x})|\psi(\mathbf{y})\rangle|^2$ for a data-dependent state $|\psi(\cdot)\rangle$ [1]. For entangling feature maps the kernel can be classically hard to evaluate, and Liu et al. [6] prove an end-to-end speed-up for a problem built on the discrete logarithm. Huang et al. [5] show that even classically hard kernels need not beat classical learners on real data, and give a geometric test for when they can. Amplitude encoding, which loads a normalized vector into the amplitudes of $\lceil \log_2 d \rceil$ qubits, is the opposite case: its kernel is classically easy, so any benefit it has must come from its inductive bias. Classical random-feature approximations [15] and Nyström methods [16] make any such kernel usable at scale. Adversarial vulnerability of quantum classifiers is studied by Lu et al. [17].

B. Machine learning for QKD attack detection

Countermeasures to implementation attacks include hardware monitoring of detector statistics [18] and protocol changes such as decoy states [19] and measurement-device-independent QKD [20]. Mao et al. [21] use neural networks to recognise attacks on continuous-variable QKD; Ren et al. [22] use machine learning for protocol and parameter selection. Such studies are almost always run on simulated telemetry, which raises the question of whether a simulator is hard enough to discriminate between methods (Section V-A).

C. Network intrusion detection

Machine-learning intrusion detection is surveyed by Buczak and Guven [23]; Sommer and Paxson [24] caution that detectors which perform well on a closed set of attacks often fail on new ones, which is why we evaluate generalization to unseen attack families directly. Kitsune [25] popularised unsupervised autoencoder detectors, which never see attack labels and so form a natural zero-day reference. Zero-trust architectures [11], [26], [27] make continuous flow-level monitoring part of the access decision.

D. Calibration and certified robustness

Security operators act on alert confidence, so probability calibration matters [28]. Gaussian processes [29] and other Bayesian models [30] provide model-based uncertainty; Platt scaling [31] and isotonic regression [32] recalibrate point estimators. Randomized smoothing [33] certifies ℓ_2 robustness for any base classifier.

III. The Amplitude-Encoding Kernel

For $\mathbf{x} \in \mathbb{R}^d$, amplitude encoding zero-pads $\mathbf{x}$ to 2^q entries, $q = \lceil \log_2 d \rceil$, normalizes it, and prepares $|\psi(\mathbf{x})\rangle = \sum_i \hat{x}_i |i\rangle$ with $\hat{\mathbf{x}} = \mathbf{x}/\|\mathbf{x}\|_2$.

Proposition 1 (Exact classical form)**.** *The fidelity kernel of amplitude encoding is*

$$K_A(\mathbf{x}, \mathbf{y}) = |\langle\psi(\mathbf{x})|\psi(\mathbf{y})\rangle|^2 = \frac{(\mathbf{x}^\mathsf{T}\mathbf{y})^2}{\|\mathbf{x}\|_2^2 \, \|\mathbf{y}\|_2^2}, \qquad (1)$$

computable in $\mathcal{O}(d)$ time per pair.

Proof. The amplitudes are real and the padding entries are zero, so $\langle\psi(\mathbf{x})|\psi(\mathbf{y})\rangle = \hat{\mathbf{x}}^\mathsf{T}\hat{\mathbf{y}}$. □

K_A is the homogeneous degree-2 polynomial kernel applied to normalized inputs, so its feature space is spanned by the products $\hat{x}_i\hat{x}_j$. Two invariances follow directly and drive the results below.

Corollary 1 (Norm and sign invariance)**.** $K_A(c\,\mathbf{x}, \mathbf{y}) = K_A(\mathbf{x}, \mathbf{y})$ *for every $c \neq 0$. Any classifier built on K_A assigns the same score to $\mathbf{x}$, $2\mathbf{x}$ and $-\mathbf{x}$; it cannot use the distance of an input from the origin.*

On standardized features the origin is the mean of the training data, so the distance from the origin is exactly the quantity an outlier detector would use. Corollary 1 therefore predicts that K_A helps when labels depend on direction and the norm is a nuisance, and hurts when an attack is recognisable mainly by being *far* from normal traffic, which is the typical situation for an attack never seen in training.

Norm anchoring. A standard way to retain the norm is to append a constant coordinate before encoding:

$$\phi_c(\mathbf{x}) = \frac{(\mathbf{x},\, c)}{\sqrt{\|\mathbf{x}\|_2^2 + c^2}}, \qquad c = \mathrm{median}_{\mathbf{x}\in\mathcal{D}_{\mathrm{train}}} \|\mathbf{x}\|_2. \quad (2)$$

The last amplitude, $c/\sqrt{\|\mathbf{x}\|^2 + c^2}$, is a monotone function of $\|\mathbf{x}\|$, and the sign of $\mathbf{x}$ is no longer ambiguous. The median-norm rule fixes c from training data alone. The encoding needs one extra amplitude (at most one extra qubit).

Sensitivity. The Jacobian of $\mathbf{x} \mapsto \hat{\mathbf{x}}$ is $(I - \hat{\mathbf{x}}\hat{\mathbf{x}}^\mathsf{T})/\|\mathbf{x}\|_2$, with spectral norm exactly $1/\|\mathbf{x}\|_2$. Normalization contracts perturbations of large-norm inputs but amplifies those of inputs near the origin, so it provides no uniform robustness guarantee; this matches the certified-robustness results in Section VI.

Implementation check. We evaluate Eq. (1) in closed form and verify it against a PennyLane [34] state-vector simulation of the overlap circuit $\langle 0|U^\dagger(\mathbf{y})U(\mathbf{x})|0\rangle$. The two agree to 7×10^{-15}; the simulated circuit takes 26 ms per kernel entry against 3×10^{-4} ms in closed form, a factor of about 8×10^4. Training-set limits of a few dozen samples that are sometimes attributed to quantum kernels are therefore a property of circuit simulation, not of this kernel.

IV. Evaluation Protocol

Matched scale. Every model in a comparison is trained on the same training subsample, at every training size n we report; feature dimensionality is matched the same way. Confidence intervals are 95% intervals over seeds (10 for the QKD and synthetic-flow benchmarks, 5 for the public datasets, 3 for leave-one-family-out runs on public data).

Models. AE-SVM uses K_A; AE-SVM (anchored) uses $K_A \circ \phi_c$. The classical kernels are RBF ($\gamma = 1/(d\,\mathrm{Var}[\mathbf{x}])$), Laplacian and cubic polynomial, all in the same SVM. Exact kernel SVMs are used up to 1,500 training samples and a Nyström approximation [16] with m uniformly sampled landmarks above that ($m = \lceil 0.1n \rceil \leq 500$ on synthetic flows, $m = 1000$ on public data); RBF, Laplacian and polynomial

kernels use the same wrapper. Other baselines are random forests (RF, 200 trees) [35], XGBoost [36], a two-layer MLP, and, on public data, a 3×256 deep network (DNN) with batch normalization, dropout and early stopping, and a Kitsune-style autoencoder [25] trained on benign traffic only. On QKD telemetry we also include a Laplace Gaussian-process classifier (GP) [29], a GP+AE-SVM stacking model [37], isotonic-calibrated RF, and three novelty detectors fit on normal data (one-class SVM, isolation forest, LOF).

Measurements. In-distribution detection (accuracy, F1, MCC, ROC-AUC); learning curves at matched n; leave-one-attack-out generalization, training on all but one attack type or family and testing on benign traffic plus the held-out one (ROC-AUC, which does not depend on a threshold tuned to known attacks); kernel-target alignment [38]; expected calibration error (ECE, 15 bins) and selective prediction; randomized-smoothing certified radii [33]; and inference latency with training excluded. Paired comparisons use paired t-tests over seeds or exact McNemar tests on test predictions.

V. Datasets

A. QKD telemetry

Each sample is one key-generation block of a decoy-state BB84 link over 50 km of fiber, with Poisson-source gain and QBER [19]. Normal operation includes per-block drift in attenuation, detector efficiency, misalignment, dark counts and background light, and finite-size noise on the two estimated error rates. Seven measurements are reported per block. Four attacks perturb them; every perturbation is scaled by a strength s, so $s = 1$ is the default signature and $s \to 0$ makes an attack indistinguishable from normal operation. Features are the 7 measurements, standardized, plus a robust Mahalanobis distance [39] to training-split normal blocks; the Mahalanobis coordinate turns out to matter (Section VI-C). We report $s = 1$ and the harder $s = 0.3$; $N = 2000$ blocks with 30% attacks.

Without drift this kind of simulator is trivially separable: in our own first version, a single feature (the error rate) separated attacks perfectly, and every model scored near 1.0. We therefore release leakage tests (independent seeds; no single-feature label proxy) with the simulator.

B. Synthetic zero-trust flows

Flow records are d-dimensional feature vectors (default $d = 64$: 16 header, 24 flow and 24 behavioral features) whose marginals follow commonly reported NetFlow statistics (Pareto packet rates, log-normal byte rates, Weibull durations) and are coupled by a Gaussian copula with within-group correlation 0.35. Five attacks (port scan, DDoS, exfiltration, DNS tunneling, ARP spoofing) shift and inflate the variance of 6–7 attack-specific latent dimensions by 1.4–2.6σ. $N = 8000$ flows with 20% attacks; features are standardized.

C. Public benchmarks

UNSW-NB15 [13]: the official partition (175,341 training and 82,332 test records) with 39 numeric features and one-hot protocol, service and state (68 features); nine attack families. Its test partition is known to be distributionally shifted from its training partition. **CIC-IDS2017** [14]: the eight MachineLearningCVE day files (2,830,743 flows). We drop rows with missing or infinite values, remove 308,381 exact duplicates and constant columns (69 features remain), drop classes with fewer than 100 flows, group labels into six families (DoS, DDoS, PortScan, BruteForce, WebAttack, Bot) and use a fixed stratified 70/30 split. Both: features are mapped through $\mathrm{sign}(x)\log(1 + |x|)$, and each seed draws a family-stratified training subsample of 20,000 flows and a test subsample of 50,000 flows.

VI. Results

A. In-distribution detection

Table I summarises in-distribution detection. In all four domains AE-SVM is statistically indistinguishable from, or slightly worse than, RBF-SVM trained on the same data: QKD MCC 0.814 vs. 0.810 (exact McNemar $p > 0.05$ on all 10 seeds); synthetic-flow accuracy 0.947 vs. 0.948 (paired t-test $p = 0.54$); UNSW-NB15 accuracy 0.862 vs. 0.863 ($p = 0.24$); CIC-IDS2017 accuracy 0.993 vs. 0.994 ($p = 0.001$, favoring RBF). Tree ensembles are the strongest family overall: RF leads on QKD telemetry, XGBoost on CIC-IDS2017, and the DNN has the highest accuracy on UNSW-NB15 (0.895), whose shifted official test partition caps every model's accuracy below 0.90 while AUCs remain ≥ 0.97. The benign-only autoencoder, which never sees an attack label, trails the supervised models in-distribution (AUC 0.887 and 0.948).

Norm anchoring does not cost in-distribution accuracy and on synthetic flows it helps: AE-SVM (anchored) reaches 0.960 accuracy and 0.983 AUC, significantly above both plain AE-SVM and RBF-SVM ($p < 10^{-4}$). On the public datasets anchoring leaves in-distribution accuracy essentially unchanged (UNSW-NB15 0.859 vs. 0.862; CIC-IDS2017 0.992 vs. 0.993).

B. Matched training size

Training every model on the same subsample (Fig. 1) shows no training size at which the amplitude kernel is reliably ahead. On QKD telemetry it needs *more* data than RBF to reach the same MCC: at $s = 0.3$ it trails by 0.03–0.04 for $n = 50$–200 and reaches parity only from $n = 1000$. On synthetic flows it trails by 3.6 points at $n = 50$ and reaches parity by $n = 1000$. On the public datasets AE-SVM tracks RBF-SVM on UNSW-NB15 at every n (AUC 0.909 vs. 0.912 at $n = 200$) and trails it on CIC-IDS2017 at small n (0.974 vs. 0.982); the Laplacian kernel is the strongest kernel at every n on both. Random forests are the most sample-efficient model on QKD telemetry (MCC 0.75 at $n = 25$, $s = 0.3$, against 0.65–0.68 for the kernels).

C. Generalization to unseen attack types

Fig. 2 shows that unseen-attack behavior differs sharply by domain. On synthetic flows, plain AE-SVM fails: its mean held-out AUC is 0.526 against 0.839 for RBF-SVM, with

TABLE I
IN-DISTRIBUTION DETECTION (MEAN ± 95% CI OVER SEEDS). QKD: MCC AND AUC AT ATTACK STRENGTH $s = 0.3$, $n = 1400$, 10 SEEDS. SYNTHETIC FLOWS: ACCURACY AND AUC, $n = 5600$, 10 SEEDS. PUBLIC DATA: F1 AND AUC ON 50,000 TEST FLOWS, $n = 20{,}000$, 5 SEEDS. – = NOT RUN IN THAT DOMAIN.

	QKD telemetry (s=0.3)		Synthetic flows		UNSW-NB15		CIC-IDS2017	
Model	MCC	AUC	Acc.	AUC	F1	AUC	F1	AUC
AE-SVM	.814±.023	**.970±.005**	.947±.002	.966±.003	.887±.003	.973±.001	.979±.001	.999±.000
AE-SVM (anchored)	.816±.019	.965±.007	.960±.003	.983±.002	.884±.003	.972±.001	.977±.001	.998±.000
RBF-SVM	.810±.015	.957±.007	.948±.003	.976±.003	.887±.003	.973±.001	.983±.001	.999±.000
Laplacian-SVM	–	–	.946±.003	.976±.002	.899±.004	.978±.001	.987±.001	**.999±.000**
Poly-SVM	–	–	.959±.003	.979±.003	.889±.003	.975±.001	.989±.001	.999±.000
GP	.801±.016	.956±.007	–	–	–	–	–	–
Hybrid	.833±.016	.969±.006	–	–	–	–	–	–
RF	.834±.013	.969±.006	.943±.003	.976±.002	.889±.003	.980±.001	.990±.002	**.999±.000**
RF-isotonic	**.837±.013**	.967±.006	–	–	–	–	–	–
XGBoost	–	–	.962±.003	**.985±.002**	.892±.003	**.981±.001**	**.994±.001**	**1.000±.000**
MLP	–	–	**.965±.001**	.984±.002	.889±.004	.975±.001	.991±.001	**.999±.000**
DNN	–	–	–	–	**.909±.006**	.977±.002	.989±.002	**.999±.000**
Autoencoder	–	–	–	–	.394±.187	.887±.009	.546±.058	.948±.012

every held-out type between 0.45 and 0.61. On QKD telemetry it generalizes as well as RF and better than RBF-SVM (mean AUC 0.952 vs. 0.917 at $s = 0.3$), but only because the feature map contains a Mahalanobis distance to normal operation, which is a magnitude: removing that one feature drops AE-SVM to 0.554 (0.635 at $s = 1$) while RBF-SVM is unaffected (0.940, 0.987).

The public benchmarks do not show the failure. On the public benchmarks, where every family is held out in turn, mean held-out AUC for AE-SVM versus RBF-SVM is: UNSW-NB15 0.945 vs. 0.932; CIC-IDS2017 0.715 vs. 0.658. On UNSW-NB15, tree ensembles and the DNN reach 0.949–0.963 and the benign-only autoencoder 0.860. On CIC-IDS2017, tree ensembles and the DNN reach 0.694–0.852 and the benign-only autoencoder 0.874. On CIC-IDS2017 every supervised model struggles on some held-out families (Bot, BruteForce, PortScan), and the benign-only autoencoder, which never sees any attack, is the most robust detector, as the closed-world critique of supervised intrusion detection [24] would predict.

D. Mechanism

Corollary 1 offers two explanations for the controlled-setting failures: sign ambiguity ($\mathbf{x}$ and $-\mathbf{x}$ look identical) and norm invariance. Interventions separate them. Mapping inputs to $[0, 1]^d$ removes the sign ambiguity but keeps norm invariance; it raises synthetic-flow zero-day AUC only from 0.526 to 0.593. The norm anchor of Eq. (2) restores the norm; it raises the same number to 0.793, and restores QKD zero-day AUC without the Mahalanobis feature to 0.939 ($s = 0.3$) and 0.980 ($s = 1$), level with RBF-SVM. On the public benchmarks the anchored encoding gives mean held-out AUC 0.940 on UNSW-NB15 (plain 0.945, non-negative 0.935); 0.733 on CIC-IDS2017 (plain 0.715, non-negative 0.663). The constant c is set by the median-norm rule from training data; a sweep on synthetic flows (two seeds) gives mean zero-day AUC 0.57, 0.80, 0.77, 0.65 and 0.53 for $c = 1, 4, 8, 16, 32$; the median training norm is 6.4, so the rule lands near the best value without looking at held-out attacks.

Can one tell in advance which regime a dataset is in? We tried the obvious statistic: for each held-out attack, the AUC of the input norm alone against the AUC of a direction-only score. Averaged per domain it is highest where AE-SVM fails (synthetic flows, norm AUC 0.90 vs. direction AUC 0.66), but across the 28 held-out splits it does not predict the AE-SVM−RBF-SVM gap (Spearman $\rho = -0.09$, $p = 0.65$), and it cannot see the QKD ablation, where removing one coordinate barely changes the norm's separating power yet flips the outcome. Which features carry magnitude matters more than how much separation the norm offers overall. We therefore do not have a reliable a-priori test; the interventions establish the mechanism where the failure occurs, and a direct leave-one-attack-out evaluation remains necessary.

Two further checks support this account. On control tasks (Fig. 3, left) whose label depends only on direction, with a heavy-tailed nuisance norm, AE-SVM learns from far fewer samples than any baseline (0.71 vs. 0.56 for RBF-SVM at $n = 25$; 0.97 vs. 0.74 at $n = 1000$); when the label depends only on the norm, it is at chance at every n. Its inductive bias is therefore real and predictable; whether it helps on a security dataset depends on how much of the attack signal lies in the norm. And kernel-target alignment [38] is lower for K_A than for RBF in every domain and attack type (synthetic flows 0.13 vs. 0.21; QKD 0.13–0.22 vs. 0.29–0.66; UNSW-NB15 0.38 vs. 0.52; CIC-IDS2017 0.62 vs. 0.67), so alignment correctly predicts in-distribution ranking but not, on its own, zero-day behaviour, which depends on which invariances the kernel imposes.

E. Calibration and selective prediction

A Gaussian-process posterior is often proposed as the uncertainty signal for security monitoring. On QKD telemetry the Laplace GP is the least calibrated supervised model (ECE 0.047 ± 0.006 at $s = 0.3$), and isotonic-calibrated RF the best (0.030). For deferral (Fig. 4) the GP's probabilities sit close to the boundary: at $u < 0.3$ it keeps 69% of blocks at 97.6% accuracy, against 80% for isotonic RF and 79% for AE-SVM at the same accuracy; at $u < 0.1$ it keeps none. The

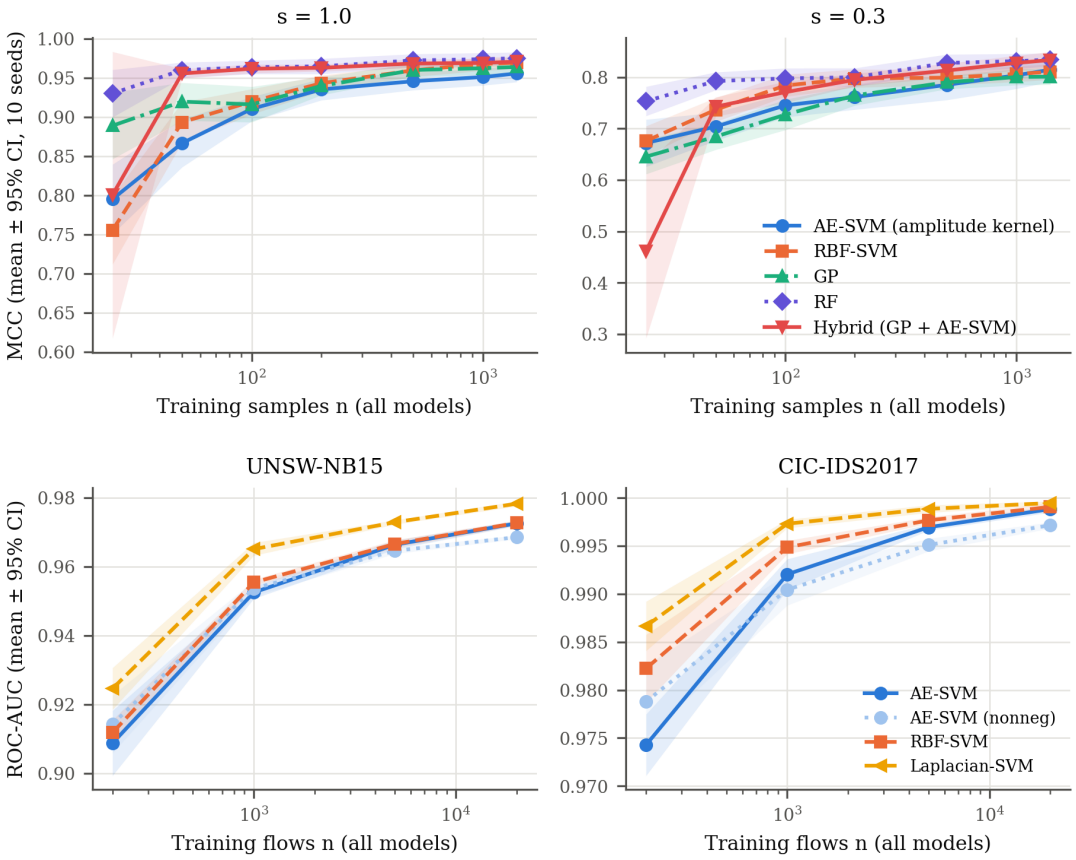


Fig. 1. Learning curves with every model trained on the same subsample at each n (mean $\pm$ 95% CI). Top: QKD telemetry, MCC, at $s = 1$ and $s = 0.3$. Bottom: public datasets, ROC-AUC.

GP's uncertainty also barely rises under a gradient-free evasion attack ($0.30 \rightarrow 0.33$), so it does not flag evasion either. Platt-scaled AE-SVM on synthetic flows supports a similar trade-off (97.4% accuracy at 75.5% coverage).

F. Certified robustness and latency

Randomized smoothing ($\sigma = 0.25$, 300 noise draws, $\alpha = 0.05$) gives comparable certified ℓ_2 radii for AE-SVM and RBF-SVM on synthetic flows (mean 0.508 vs. 0.518; median radius saturated at 0.582 for both), with AE-SVM abstaining on 1.8% of points versus none for RBF-SVM. This agrees with the sensitivity analysis of Section III: normalization does not confer uniform robustness. All detectors are fast. In batch inference AE-SVM costs 0.008 ms per QKD block and 0.0048 ms per synthetic flow with 50 Nyström landmarks, about $1.2\times$ faster than RBF-SVM, because it avoids an exponential per landmark; tree ensembles and the GP are slower but still far below any per-flow budget. On the public data, however, kernel SVMs with 1000 Nyström landmarks are the slowest models in batch inference (0.23–0.38 ms per flow, averaged over both datasets) while XGBoost, the MLP and the autoencoder need 0.002–0.020 ms; these timings come from a shared, heavily loaded machine and should be read as an ordering only.

VII. Discussion

A. What the results say about quantum kernels in security

Our evidence concerns one encoding, but it is the encoding most often used. Amplitude encoding brings no in-distribution advantage over an RBF kernel trained on the same data in any of four datasets, and at small n it needs more data, not less. Its distinctive property is an invariance, not a capacity: it sees direction and discards magnitude. That is an advantage when magnitude is a nuisance, as in our direction-only control task, and a liability when unseen attacks differ from normal traffic mainly in magnitude, as in both of our controlled settings. On real traffic, where attack families also differ in *which* features are active, the liability did not materialize, and the kernel generalized slightly better than RBF. Two further points. (i) Nothing here bears on quantum speed-ups: K_A is classically easy, and the slow circuit simulation that limits some published comparisons to a few dozen samples is an artifact. (ii) The invariance analysis applies to any fidelity kernel whose state preparation normalizes its input. A third, methodological point is that controlled simulators alone would have led us to the wrong general conclusion; the public benchmarks reversed it.

B. Recommendations

For practitioners evaluating quantum or quantum-inspired kernels for security detection we recommend: (1) compare

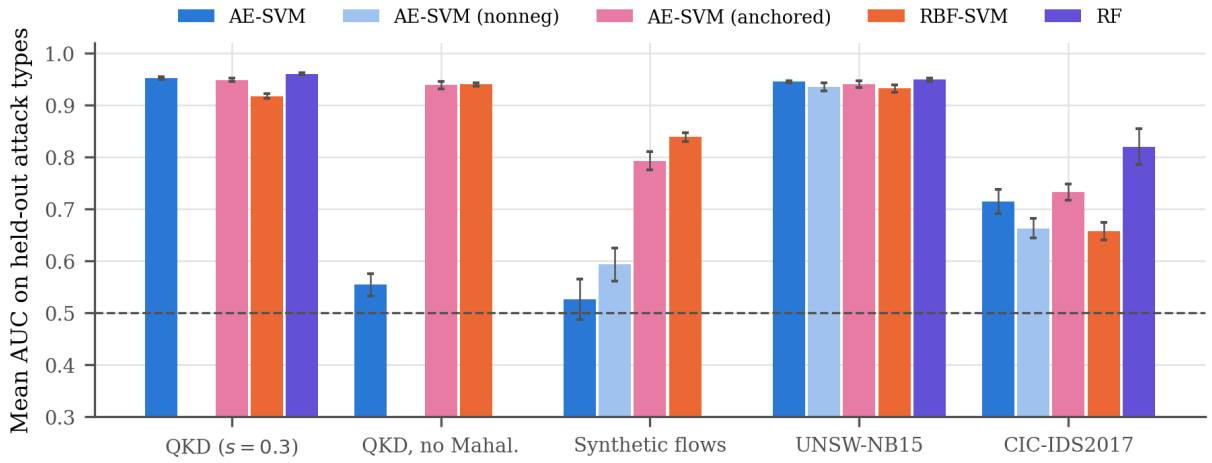


Fig. 2. Mean ROC-AUC on held-out attack types (leave one type or family out; mean ± 95% CI over seeds). "QKD, no Mahal." removes the Mahalanobis feature, the only QKD feature that carries the input's magnitude.

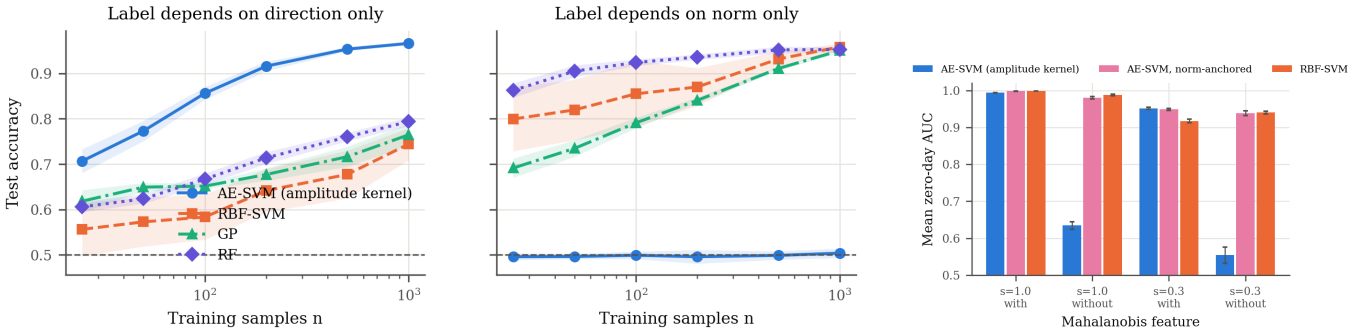


Fig. 3. Left pair: control tasks in $\mathbb{R}^8$ with heavy-tailed norms; the label depends on direction only or on norm only. Right: QKD zero-day AUC with and without the Mahalanobis feature.

against classical kernels trained on the *same* data at every n; (2) report leave-one-attack-out results on real traffic, since in-distribution accuracy hid the largest effect in this study and simulators alone pointed to the wrong general conclusion; (3) check which input transformations the kernel is invariant to, and whether any feature re-injects what it discards; (4) if amplitude encoding is used, anchor the norm (Eq. (2)), which repaired every failure we observed and lowered no public-benchmark held-out AUC by more than 0.005; and (5) for simulated telemetry, report single-feature separability and a signature-strength sweep, because a simulator without operating drift can make any detector look perfect.

C. Threats to validity

Simulation. QKD results come from a simulator whose attack magnitudes are modeling assumptions; the strength axis shows how conclusions move with signature size but does not fix a realistic s. Synthetic flows are a marginal-plus-copula model, not packet-level traffic. The public datasets mitigate this for network flows but not for QKD, for which no public attack telemetry exists to our knowledge. *Datasets.* UNSW-NB15 and CIC-IDS2017 have known labeling and distribution artifacts; we use the official UNSW-NB15 partition, de-duplicate CIC-IDS2017, and subsample to 20,000 training flows per seed, which favors kernel methods relative to full-data tree ensembles. *Models.* Kernel hyperparameters ($C = 1$, median-heuristic bandwidths) were not tuned per dataset, for any kernel; tuning could change small gaps but is unlikely to close the 0.3-AUC zero-day gaps in the controlled settings. Nyström approximation with 1000 landmarks may understate exact-kernel accuracy equally for all kernels. *Zero-day protocol.* Leaving out one attack family is a proxy for novel attacks; a real novel attack may be closer to, or farther from, the training families. *Robustness.* Certified radii are for ℓ_2 noise on standardized features and do not cover semantic or adaptive evasion.

VIII. Conclusion

We evaluated the amplitude-encoding quantum kernel—exactly, at every training size, against classical kernels trained on the same data—on QKD telemetry, synthetic zero-trust flows, UNSW-NB15 and CIC-IDS2017. It never beats an RBF kernel in distribution and needs more data at small n. On unseen attack types it fails in both controlled settings, because its fidelity kernel discards input norm, and a one-coordinate norm anchor repairs that failure; on both public

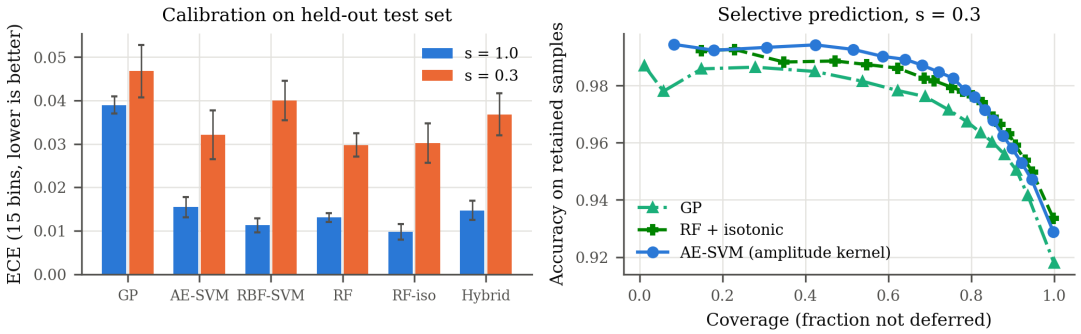


Fig. 4. QKD telemetry. Left: ECE by model. Right: accuracy versus coverage when deferring samples whose uncertainty $u = \sqrt{p(1-p)}$ exceeds a threshold ($s = 0.3$).

benchmarks it does not fail, and matches or slightly exceeds RBF. Since a simple statistic did not predict which case applies, quantum and quantum-inspired detectors should be judged under a matched-scale protocol that includes held-out attack families on real traffic, with their invariances analysed before deployment.

Code and Data Availability

All code, the two simulators, dataset preparation scripts for the public benchmarks, experiment drivers and every result file behind the tables and figures are available at https://github.com/arnavk23/quasar-security.

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